

Leading Kazhdan–Lusztig coefficients with fully commutative lower endpoints in type E

Abstract

We study the leading coefficients $\mu(x, w)$ of ordinary equal-parameter Kazhdan–Lusztig polynomials when x is fully commutative. For finite Weyl groups of types E_6 , E_7 , and E_8 , we prove $\mu(x, w) \in \{0, 1\}$ for every upper endpoint w . The proof combines star reduction with exhaustive classifications of terminal elements; parabolic pruning avoids full-group enumeration in type E_8 . Together with the known type- A and type- D results, this establishes the bound for every finite simply laced Coxeter group. We also construct full-support terminal reflections in every generalized type E_{4r+1} , $r \geq 3$. Green’s monomial cell theory implies that their leading coefficients with fully commutative lower endpoint vanish away from covers and, at each fixed rank, vanish for all sufficiently large values of the family parameter. An infinite family in affine type $\tilde{E}_8 = E_9$ has every such leading coefficient equal to zero. A complementary E_{10} construction is recorded in an appendix.

1 Introduction

Let (W, S) be a Coxeter system, with length ℓ and Bruhat order $\leq$. Write $P_{x,w}(q)$ for its ordinary equal-parameter Kazhdan–Lusztig polynomials, and define

$$\mu(x, w) = \begin{cases} [q^{(\ell(w)-\ell(x)-1)/2}]P_{x,w}(q), & x < w \text{ and } \ell(w) - \ell(x) \text{ is odd,} \\ 0, & \text{otherwise.} \end{cases}$$

An element is fully commutative if any two of its reduced expressions are related by interchanging adjacent commuting generators. Denote this set by W_{FC} . The first result concerns all upper endpoints in the finite exceptional groups.

Theorem 1 (Finite exceptional types). *If W has type E_6 , E_7 , or E_8 , then*

$$x \in W_{\text{FC}}, \quad w \in W \quad \implies \quad \mu(x, w) \in \{0, 1\}.$$

The proof extends Gern’s type- D star argument [2, Theorem 4.5.11]. It reduces to a finite classification of upper endpoints with no noncommuting initial or final pair in any reduced expression. We call these elements *terminal*; a terminal is *noncommuting* if it is not a product of mutually commuting generators. There are respectively 1, 4, and 6 noncommuting terminals in E_6 , E_7 , and E_8 . Each has a unique fully commutative lower endpoint below it satisfying both descent inclusions. All resulting coefficients vanish by parity except for a known pair in a D_6 parabolic. The E_8 classification uses exhaustive parabolic pruning rather than enumeration of its 696,729,600 elements.

Corollary 2 (Finite simply laced groups). *If W is a finite simply laced Coxeter group, $x \in W_{\text{FC}}$, and $w \in W$, then $\mu(x, w) \in \{0, 1\}$.*

The irreducible type- A and type- D cases are due to Green [5] and Gern [2, Theorem 4.5.11]. Theorem 1 supplies the exceptional cases; the direct-product argument is given in Section 3.

Our second result applies in arbitrarily large ranks. Throughout, the generalized diagram E_n , $n \geq 6$, is the chain $0 - 1 - \cdots - (n-2)$ with node $n-1$ attached to node 2. Thus $E_9 = \tilde{E}_8$, and E_n is indefinite for $n \geq 10$. The numbering is

$$\begin{array}{ccccccc} & & & n-1 & & & \\ & & & | & & & \\ 0 & - & 1 & - & 2 & - \cdots - & n-2 \end{array}$$

The *support* of an element is the set of generators occurring in its reduced expressions; full support means $\text{supp}(w) = S$.

Theorem 3 (A rank-uniform family). *For every integer $r \geq 3$ there is an explicit family of distinct full-support, non-fully-commutative terminal reflections $b_{r,k} \in W(E_{4r+1})$, $k \geq 0$. For every $x \in W(E_{4r+1})_{\text{FC}}$,*

$$\mu(x, b_{r,k}) \in \{0, 1\}, \quad x < b_{r,k}, \quad \ell(b_{r,k}) - \ell(x) > 1 \implies \mu(x, b_{r,k}) = 0.$$

For each fixed r , all these leading coefficients vanish for every sufficiently large k .

These families show that, in the indicated infinite groups, the terminal obstructions to length-decreasing star reduction cannot be exhausted by a finite list. The construction in Section 4 starts from a staircase root and iterates an affine-parabolic root action. An even sign change in D_{4r} proves real-root membership uniformly in r . The common descent set is an independent set of maximum cardinality and has odd size. Green’s monomial Temperley–Lieb cell theorem then restricts a potentially nonzero non-cover coefficient to a single fully commutative lower endpoint, and parity forces that coefficient to vanish. This proof requires no Kazhdan–Lusztig polynomial computation.

Green proved the fully commutative lower-endpoint bound in type A and affine type A [5]; Chmutov proved it in finite type B [14, §2.3, Theorem 2.3.8], hence also for the identical Coxeter system of type C . The two-FC-endpoint bound in generalized type E_n goes back to Graham’s thesis, as recalled by Green [11, Remark 9.5]. Green’s trace formulas [6, Remark 7.12] and Jones’s result with Deodhar upper endpoint [15, Theorem 1.2] have different endpoint hypotheses. The structural input for both the finite lower-endpoint analysis and the infinite families is Green’s cell theory in generalized type E [11]. Our contribution is a complete certified exceptional terminal classification and its coefficient consequence, together with the explicit rank-uniform terminal construction. The descent and parity mechanism is classical. We are not aware of an earlier proof of the exceptional bound or an earlier description of these terminal lists and rank-uniform families; historical priority remains unconfirmed.

Exceptional KL computations already feature in Geck’s *PyCox* [8] and Geck–Halls’s type- E_8 cell analysis [9]. Our assertion concerns every coefficient $\mu(x, w)$ with FC lower endpoint, rather than only cell partitions or leading coefficients of character values. A direct appeal to FC cell closure is unavailable in type E : Green–Losonczy’s Example 2.2.5 [7] shows that this closure property fails there. Moreover, a left W -graph action records a coefficient for a pair (x, w) only through a generator in $L(x) \setminus L(w)$ [9, Remark 2.1]. Our residual terminal pairs have equal left and right descent sets, so the cell action matrices alone do not determine their coefficients.

The infinite-type results concern specified upper families, rather than arbitrary upper endpoints. They control μ , not all coefficients of $P_{x,w}$. Section 5 gives the complementary affine E_8 family; the rank-ten supplement is in Appendix C. Section 6 records a conditional consequence of combinatorial invariance. Exact computational certificates and reproduction instructions appear in Appendix A.

2 Structural reductions

Write

$$L(w) = \{s \in S : \ell(sw) < \ell(w)\}, \quad R(w) = \{s \in S : \ell(ws) < \ell(w)\}.$$

An independent vertex set in the Coxeter graph is a set of mutually commuting simple generators. For such a set I , write $i(I) = \prod_{s \in I} s$; the order of the factors is immaterial. The terminal condition is Gern's "weakly bad" condition and Laird's T -avoiding condition [2, 3]; a terminal which is not a commuting product is called "bad" in [2]. Laird listed the type- E_n classification as a future direction in 2016 [3, §5.3]. We classify the finite exceptional cases and give infinite families, without asserting a classification in infinite type.

2.1 Star reduction

We use the descent criterion and length-three star transport [1, §2.3 and Theorem 4.2], also stated in [2, Proposition 1.2.4]. Full commutativity is preserved by these stars in simply laced systems [4, Proposition 2.10]; see also [2, Proposition 1.1.29].

Lemma 4 (Terminal reduction). *Let W be simply laced. If $\mu(x, b) \in \{0, 1\}$ for every terminal upper endpoint b and every fully commutative x , then the same conclusion holds for all upper endpoints in W .*

Proof. Induct on $\ell(w)$ and put $d = \ell(w) - \ell(x)$. Noncomparable pairs give zero. For comparable pairs, even length difference gives zero and difference one gives one. Hence assume $x < w$ and $d \geq 3$ is odd. If $L(w) \not\subseteq L(x)$ or $R(w) \not\subseteq R(x)$, the descent criterion gives zero: its only nonzero exception is a cover, already excluded. We can thus assume both inclusions. Since the descent sets of a fully commutative element are pairwise commuting, so are $L(w)$ and $R(w)$.

If w is terminal, use the hypothesis. Otherwise, suppose first that $w = stv$ is reduced with $m(s, t) = 3$. Then $s \in L(w)$ and $t \notin L(w)$. The left $\{s, t\}$ -star is $*w = sw = tv$, with $\ell(*w) = \ell(w) - 1$. Descent inclusion gives $s \in L(x)$, while full commutativity gives $t \notin L(x)$. Therefore $*x$ is defined and fully commutative. Star transport gives

$$\mu(x, w) = \tilde{\mu}(*x, *w),$$

where $\tilde{\mu}$ is symmetric and zero on incomparable pairs. The new length difference is d or $d - 2$, so $\ell(*x) < \ell(*w)$. If the new endpoints are incomparable the value is zero. Otherwise the induction hypothesis applies to the shorter upper endpoint $*w$. The case of a noncommuting suffix uses right stars in the same way. $\square$

2.2 Maximum commuting descents

The monomial cell structure is due to Fan [10]; we use Green's formulation [11, §4]. The next lemma makes explicit the singleton-cell argument in the final paragraph of Green's proof of Lemma 5.2 [11]. It supplies the lower-endpoint analysis in both the finite and infinite cases.

Lemma 5 (Maximum common commuting descents). *Let x be fully commutative in E_n , and let K contain its support. Let I be an independent set of maximum cardinality in the induced graph on K . If $I \subseteq L(x) \cap R(x)$, then $x = i(I)$.*

Proof. Use Green's monomial Temperley–Lieb function a_{TL} , the maximum size of a commuting product occurring as a length-additive factor [11, Definition 4.1]. The descent inclusions give

both a prefix and a suffix $i(I)$. Thus $a_{\text{TL}}(x) \geq |I|$, whereas the support and maximum-cardinality hypothesis give $a_{\text{TL}}(x) \leq |I|$. Green's Proposition 4.4(iv), and its left-hand version, put x in both the monomial right and left cells of $i(I)$. Proposition 4.4(v) says that this intersection is a singleton, already containing $i(I)$. Therefore $x = i(I)$. No identification of these monomial cells with ordinary KL cells is used. $\square$

Proposition 6 (Maximum-descent parity criterion). *Let $b \in W(E_n)$ satisfy $L(b) = R(b) = I$, where I is a maximum independent set in the subgraph induced by $\text{supp}(b)$. If $\ell(b) \equiv |I| \pmod{2}$, then for every FC x*

$$\mu(x, b) \in \{0, 1\}, \quad x < b, \ell(b) - \ell(x) > 1 \implies \mu(x, b) = 0.$$

This criterion does not require b to be terminal.

Proof. For a non-cover with a potentially nonzero coefficient, descent containment is necessary. Since $x \leq b$, its support lies in that of b . Lemma 5 forces $x = i(I)$. The resulting length difference is even, so its coefficient is zero. Covers have coefficient one. $\square$

The independence number of E_n is $\lceil n/2 \rceil$. Excluding node 2 leaves three paths with maximum total $2 + \lceil (n-4)/2 \rceil$; including it gives $2 + \lceil (n-5)/2 \rceil$. Taking the larger proves the claim.

2.3 Geometric terminal tests

Normalize simple roots by $(\alpha_j, \alpha_j) = 2$ and $(\alpha_s, \alpha_t) = -1$ for adjacent nodes. For a real root β , its reflection acts by $r_\beta(\rho) = \rho - (\rho, \beta)\beta$. Root inequalities refer to positivity in the simple-root basis.

In these coordinates, the right-terminal test is

$$s \in R(w) \implies w(\alpha_s + \alpha_t) > 0 \quad \text{for every } t \sim s. \quad (1)$$

Indeed, $(ws)(\alpha_t) = w(\alpha_s + \alpha_t)$ when $s \sim t$. Testing w^{-1} gives the left-terminal condition.

3 Finite exceptional types

3.1 The terminal classification

A fully commutative x is *eligible* for a terminal b if

$$x \leq b, \quad L(b) \subseteq L(x), \quad R(b) \subseteq R(x).$$

For non-covers, every ineligible pair has $\mu(x, b) = 0$ by the descent criterion. Words in the finite types are written as strings of node labels.

Proposition 7 (Computational classification). *The noncommuting terminals in E_6 , E_7 , and E_8 are exactly those in Table 1. Both descent sets of each terminal are the support of the commuting product x shown in the table. The remaining terminals are commuting products: 22, 36, and 58, respectively, including the identity.*

Here $b_{8,6}$ is the concatenation of the two reduced-word strings

$$(7534231270123456210321432)(5437210321432543721324357).$$

Type	Terminal b	$\ell(b)$	$x = i(I)$	$\ell(b) - \ell(x)$
E_6	1325213	7	135	4
E_7	1326213	7	136	4
E_7	13256213	8	1356	4
E_7	132543621324356	15	1356	11
E_7	1325436210321432543621324356	28	1356	24
E_8	1327213	7	137	4
E_8	13257213	8	1357	4
E_8	61327213	8	1367	4
E_8	132543721324357	15	1357	11
E_8	1325437210321432543721324357	28	1357	24
E_8	$b_{8,6}$	50	1357	46

Table 1: All noncommuting terminals and their common commuting descent products.

Corollary 8. *Each terminal in Table 1 has the unique eligible FC lower endpoint x displayed there.*

Proof. The induced support graphs are D_4 , $D_4 \sqcup A_1$, D_6 , E_7 , or E_8 . Their independence numbers are respectively 3, 4, 4, 4, 4. The common descent set in each row has that cardinality. If y is eligible, its support lies in that of b and Lemma 5 forces y to be the displayed commuting product x . Conversely, x is a commuting descent product of b , hence a reduced prefix and therefore below b in Bruhat order. $\square$

For a direct check of the independence bounds, use the matching $(1, 2)$ in the length-7 and length-8 rows, $(1, 2), (3, 4)$ in the length-15 rows, $(0, 1), (2, 3), (4, 5)$ in the length-28 rows, and $(0, 1), (3, 4), (5, 6), (2, 7)$ in the length-50 row. Each bounds an independent set by $|\text{supp}(b)|$ minus the matching size, with equality attained by the displayed descent set.

3.2 Exhaustiveness of the classification

All computations use the faithful integral geometric representation. For columns $a_j = w(\alpha_j)$, right multiplication by s_i replaces

$$a_i \mapsto -a_i, \quad a_j \mapsto a_j + a_i \quad (j \sim i),$$

and fixes the other columns. Signs determine descents, and (1) tests terminality on both sides. A noncommuting suffix ts is equivalent to $s \in R(w)$ and $t \in R(ws)$. In E_6 and E_7 , breadth-first multiplication by all generators until closure enumerates the complete group and gives minimal word lengths.

For E_8 , write the unique length-additive parabolic decomposition $w = av$, with $a \in W^J$ and $v \in W_J$, where W^J consists of the minimal right coset representatives. Call an element *right-terminal* if it has no reduced expression ending in a noncommuting pair.

Lemma 9 (Parabolic pruning). *If $w = av$ is right-terminal, then v is right-terminal in W_J .*

Proof. A reduced expression for v ending in adjacent generators would, after prefixing a reduced expression for a , give such an expression for w . Length additivity rules this out. $\square$

Consequently one may enumerate all products av with v right-terminal, apply (1) in the ambient group, and then test inverses to select the two-sided terminals. The recursive implementation follows

$$E_8 \supset E_7 \supset E_6 \supset D_5 \supset D_4 \supset A_3 \supset A_2 \supset A_1 \supset \{e\}.$$

At each step all minimal coset representatives are obtained by closure under left multiplication, reducing to right- J -minimal form by stripping right J -descents. This reaches every coset because the left action on right cosets is transitive. Starting with the identity, induction using Lemma 9 proves completeness at every stage. The final E_7 parabolic has 576 right-terminals and 240 cosets. Testing their products gives 2,160 right-terminals in E_8 ; the inverse test leaves exactly the 64 terminals in Proposition 7.

The computation certifies the terminal classification; the eligible lower endpoints follow analytically from Corollary 8. In particular, the finite theorem requires neither an enumeration of all FC-upper pairs nor a large KL polynomial table. Independent FC catalogues provide a supplementary check of the lower-endpoint assertion. Their closure test, counts, and certificate locations are given in Appendix A.

Proof of Theorem 1. By Lemma 4, it suffices to treat terminal upper endpoints. For a commuting product b , every $x < b$ is a proper subproduct. The descent criterion gives zero on non-covers and one on covers. For a noncommuting terminal, Table 1 and parity leave only the length-15 rows. These are the same pair in a D_6 parabolic. In E_n , $n = 7, 8$, mapping Gern's labels 1, 2, 3, 4, 5, 6 to $1, n-1, 2, 3, 4, 5$ identifies the eligible pair with

$$x_6 = (-1, -2, 4, 3, 6, 5), \quad w_6 = (-1, -6, 3, -4, 5, -2).$$

Bruhat order and ordinary KL polynomials agree with those in the standard parabolic. Gern's Lemma 4.5.5 [2] gives $\mu(x_6, w_6) = 1$. All other non-cover coefficients with terminal upper endpoint are zero. Lemma 4 now proves the theorem. $\square$

Proof of Corollary 2. The irreducible classification is A , D , and E ; combine Green's type- A theorem, Gern's type- D theorem, and Theorem 1. For a direct product $W = \prod_j W_j$, full commutativity is componentwise and $P_{x,w} = \prod_j P_{x_j, w_j}$. Suppose $x < w$, and let m be the number of components with positive length difference d_j and $d = \sum_j d_j$. An even d gives $\mu = 0$. For odd d and $m \geq 2$, the usual degree bound gives

$$\deg P_{x,w} \leq \sum_{d_j > 0} \frac{d_j - 1}{2} = \frac{d - m}{2} < \frac{d - 1}{2},$$

so $\mu = 0$. For $m = 1$ it is the coefficient in the single nontrivial component, which lies in $\{0, 1\}$. Noncomparable pairs give zero. $\square$

4 The rank-uniform construction

Fix an integer $r \geq 3$ and put $n = 4r + 1$.

Number the chain $0 - 1 - \dots - (4r - 1)$ and attach node $4r$ to node 2. Define a vector β_r by

$$(\beta_{r,0}, \beta_{r,1}, \beta_{r,2}, \beta_{r,4r}) = (r - 1, r, 2r - 1, r), \quad \beta_{r,j} = 2r - \lfloor j/2 \rfloor \quad (3 \leq j \leq 4r - 1).$$

Thus the long-arm coordinates are $2r - 1, 2r - 2, 2r - 2, \dots, 1, 1$. Put

$$I_r = \{0, 3, 5, \dots, 4r - 1, 4r\}, \quad a = 2r - 4.$$

Use the embedded affine E_8 subdiagram on $\{0, \dots, 7, 4r\}$ and set

$$\begin{aligned} (\delta_0, \dots, \delta_7) &= (2, 4, 6, 5, 4, 3, 2, 1), & \delta_{4r} &= 3, \\ (\gamma_0, \dots, \gamma_7) &= (1, 2, 3, 2, 2, 1, 1, 0), & \gamma_{4r} &= 1. \end{aligned}$$

All their other coordinates vanish. Set $T = r_\gamma r_{\gamma+\delta}$. For $k \geq 0$ define

$$\beta_{r,k} = \beta_r - ak\gamma + (ak^2 + rk)\delta, \quad b_{r,k} = r_{\beta_{r,k}}. \quad (2)$$

4.1 The staircase seed

Lemma 10. *For every $r \geq 3$, the vector β_r is a real root of E_{4r+1} .*

Proof. Deleting node 0 gives finite type D_{4r} . In an orthonormal basis $e_1, \dots, e_{4r}$ realize its simple roots as

$$\alpha_1 = e_1 + e_2, \quad \alpha_{4r} = e_2 - e_1, \quad \alpha_j = e_{j+1} - e_j \quad (2 \leq j \leq 4r-1).$$

Take z orthogonal to this space, with $(z, z) = 2 - r$, and put $\alpha_0 = z - \frac{1}{2} \sum_i e_i$. These vectors have exactly the generalized E_{4r+1} Cartan inner products. Expanding the displayed seed gives

$$\beta_r = (r-1)z + \sum_{i=1}^{4r} c_i e_i, \quad c_i = \begin{cases} -(r-1)/2 & i \text{ odd}, \\ (3-r)/2 & i \text{ even}. \end{cases}$$

Since $(\beta_r, \alpha_0) = r-2$, we have

$$s_0 \beta_r = z - \frac{1}{2} \sum_{i \text{ odd}} e_i + \frac{1}{2} \sum_{i \text{ even}} e_i.$$

This equals $F\alpha_0$, where F flips the $2r$ even coordinates. An even sign change belongs to $W(D_{4r})$ [2, Example 1.1.6]. Explicitly, it is a product of r double sign changes, each given by $r_{e_{4j-2}-e_{4j}} r_{e_{4j-2}+e_{4j}}$, $1 \leq j \leq r$. Therefore $\beta_r = s_0 F \alpha_0$ is a real root for every r . □

4.2 The affine-parabolic action

We record the elementary reflection calculation underlying the iteration.

Lemma 11. *Suppose that γ and $\gamma + \delta$ are real roots, with $(\gamma, \gamma) = 2$ and $(\gamma, \delta) = (\delta, \delta) = 0$. For $T = r_\gamma r_{\gamma+\delta}$,*

$$(T-1)\rho = (\rho, \delta)\gamma - ((\rho, \gamma) + (\rho, \delta))\delta, \quad (T-1)^3 = 0.$$

If $(\beta, \delta) = -a$ and $(\beta, \gamma) = -c$, then

$$T^k \beta = \beta - ak\gamma + (ak^2 + ck)\delta \quad (k \geq 0). \quad (3)$$

Proof. The first identity follows by multiplying the two reflection formulas. Set $N = T - 1$. Then $N\delta = 0$ and $N\gamma = -2\delta$, so $N^3 = 0$. Moreover $N\beta = -a\gamma + (a+c)\delta$ and $N^2\beta = 2a\delta$. The binomial expansion of $(1+N)^k$ gives (3). □

The lemma applies on the whole ambient root space. In the indefinite constructions below, δ is null in an affine parabolic but is not in the radical of the ambient form.

4.3 Translated roots and descent sets

Lemma 12. *For all $r \geq 3$ and $k \geq 0$, $\beta_{r,k} = T^k \beta_r$ is a real root with every coordinate strictly positive. Its reflection is terminal and has both descent sets equal to I_r .*

Proof. The vector γ is a root of the finite E_8 parabolic obtained by omitting node 7, and $\gamma + \delta$ is an affine real root. Fixed lowering witnesses are given in Appendix B. Lemma 11 applies to T . Direct pairing with the seed gives

$$(\beta_r, \gamma) = -r, \quad (\beta_r, \delta) = -a.$$

Equation (3) gives $T^k \beta_r = \beta_{r,k}$. Thus every vector in (2) is a real root. In the binomial form its first increment is $(a+r)\delta - a\gamma$ and its second increment is $2a\delta$. Both are coordinatewise nonnegative for $r \geq 3$, so every coordinate of $\beta_{r,k}$ is strictly positive.

Set $m_j = (\alpha_j, \beta_{r,k})$. The pairings are

$$\begin{aligned} m_0 &= r - 2, & m_1 &= 2 - r, & m_2 &= -1 - ak, \\ m_j &= (-1)^{j+1}(1 + ak) & (3 \leq j \leq 7), \\ m_8 &= -1 - ak^2 - rk, \\ m_j &= (-1)^{j+1} & (9 \leq j \leq 4r - 1), & m_{4r} &= 1 + ak. \end{aligned}$$

They are positive exactly on I_r . The formula

$$b_{r,k}(\alpha_j) = \alpha_j - m_j \beta_{r,k}$$

is negative exactly when $m_j > 0$, since $\beta_{r,k}$ is a strictly positive integral real root. Hence $R(b_{r,k}) = I_r$, and involutivity gives $L(b_{r,k}) = I_r$. For every edge incident to I_r , the sum of its two pairings is nonpositive. At the exceptional edge $7 - 8$ this sum is $-ak(k - 1) - rk \leq 0$ for integer $k \geq 0$; at $8 - 9$ it is $-ak^2 - rk$. The remaining edge sums are nonpositive by the displayed formulas. For $s \in I_r$ and $t \sim s$,

$$b_{r,k}(\alpha_s + \alpha_t) = \alpha_s + \alpha_t - (m_s + m_t) \beta_{r,k} > 0.$$

Thus deleting a right descent creates no adjacent right descent. The reflection is an involution, so the same holds on the left. □

Proof of Theorem 3. The set I_r is independent of size $2r + 1$. The matching

$$(0, 1), (2, 4r), (3, 4), (5, 6), \dots, (4r - 3, 4r - 2)$$

has $2r$ edges and leaves one vertex unmatched, proving that I_r is maximum. Every reflection has odd length, so Proposition 6 proves the coefficient assertions. If $b_{r,k}$ were FC, Lemma 5 would force it to be $i(I_r)$, whose (-1) -eigenspace has dimension $2r + 1$, whereas a real-root reflection has dimension one.

Full support follows because $(\alpha_0, \beta_{r,k}) = r - 2 > 0$ and every root coordinate is positive: column 0 of the reflection changes every coordinate row. Each simple reflection changes only its own coordinate of a vector, so an omitted generator would leave that coordinate row unchanged. Distinctness follows from

$$\beta_{r,k,0} = r - 1 + 4k + (4r - 8)k^2.$$

Distinct positive norm-two roots give distinct reflections, since the image of $1 - r\beta$ is the line spanned by β . Finally, the FC set in each fixed generalized E_n is finite [12, Theorem 5.1]. Finite Coxeter balls and distinctness imply $\ell(b_{r,k}) \rightarrow \infty$ as $k \rightarrow \infty$. For sufficiently large k , its length exceeds the maximum FC length by more than one, so cover coefficients are also excluded. □

At $r = 3$ the seed is

$$\beta_3 = (2, 3, 5, 5, 4, 4, 3, 3, 2, 2, 1, 1, 3),$$

giving the E_{13} family. The restriction $r \geq 3$ is structural: at $r = 2$, $(\alpha_0, \beta_2) = 0$ and the maximum odd descent set is lost. The argument does not give an effective threshold for the eventual vanishing in Theorem 3.

5 An affine vanishing family

Use E_9 numbering: chain $0 - 1 - \dots - 7$, with node 8 attached to 2. The affine node is 7. Coordinates are in the simple-root basis, with adjacent pairings -1 and squared simple-root lengths 2. Put

$$\begin{aligned} \delta &= (2, 4, 6, 5, 4, 3, 2, 1, 3), \\ \gamma &= (1, 1, 1, 1, 1, 0, 0, 0, 0), & I &= \{1, 3, 5, 7, 8\}, \\ b_0 &= 312875645234123012856745231. \end{aligned}$$

Let $T = r_\gamma r_{\gamma+\delta}$, so that $T(\alpha) = \alpha - (\alpha, \gamma)\delta$, and define $b_k = T^k b_0 T^{-k}$ for integers $k \geq 0$.

Theorem 13 (Explicit affine family). *Every b_k is a full-support terminal element in affine type $\tilde{E}_8$ that is not fully commutative, with*

$$L(b_k) = R(b_k) = I, \quad \ell(b_k) = 27 + 92k.$$

For every fully commutative $x \in W(E_9)$, one has $\mu(x, b_k) = 0$.

Proof. The vector δ is the null root, and γ is a root of the finite E_8 parabolic obtained by omitting node 7. Multiplying the displayed word gives $b_0^2 = 1$. Write B_0 for the matrix with columns $b_0(\alpha_j)$. The matrix is recorded in Appendix B. Its columns are negative precisely at the indices in I , and for every edge incident to I the sum of the corresponding columns is a positive root. These checks prove that b_0 is terminal on both sides. Put

$$d = (2, -1, 1, -1, 1, -1, 1, -1, -1).$$

Computing $(\alpha_j, \gamma - b_0\gamma)$ gives d_j , and conjugation yields

$$b_k(\alpha_j) = b_0(\alpha_j) + kd_j\delta. \tag{4}$$

The base columns are negative on I and positive off I ; d has the same sign pattern. Thus the descent set stays I . The involution property persists under conjugation, giving the same left descent set. On every edge $s \sim t$ with $s \in I$, one has $d_s + d_t \geq 0$. Consequently (4) preserves positivity of $b_k(\alpha_s + \alpha_t)$, which is precisely the right-terminal test. Involution gives the left test. At $k = 0$ the displayed reduced word has full support. At $k \geq 1$ each descent column is strictly negative in every coordinate. Omitting a simple generator would leave its coordinate row unchanged, contradicting these columns. Distinctness follows from the nonzero column slopes.

For completeness, count affine inversions to obtain the exact length. For each of the 240 finite roots η , write $b_0\eta = \eta' + h\delta$ and let $\epsilon(\eta)$ be 0 for positive finite roots and 1 for negative ones. The positive affine roots with finite part η are $\eta + n\delta$, $n \geq \epsilon(\eta)$. Their contribution to the length of b_k is

$$\max(0, A_\eta + kB_\eta), \quad A_\eta = \epsilon(\eta') - h - \epsilon(\eta), \quad B_\eta = -\sum_j \eta_j d_j.$$

Finite-root enumeration groups the potentially nonzero contributions by (A_η, B_η) as follows:

(A, B)	$(0, 1)$	$(0, 2)$	$(1, 0)$	$(1, 1)$	$(1, 2)$
Multiplicity	44	8	1	20	6

The remaining 161 contributions are identically zero for $k \geq 0$. The sum is $27 + 92k$. Appendix A identifies the complete root-string certificate.

The set I is a maximum independent set of size 5. Proposition 6 therefore gives zero on all FC non-covers. The complete FC catalogue has maximum length 44, so there are no FC covers when $k \geq 1$. For $k = 0$, delete one letter from the reduced word and retain products of length 26. This gives all 21 distinct Bruhat covers, none FC. It eliminates the cover exception. If b_k were FC, Lemma 5 would force $b_k = i(I)$, contradicting $\ell(b_k) = 27 + 92k > 5$. $\square$

The first indefinite rank E_{10} is outside the uniform sequence E_{4r+1} , $r \geq 3$. Appendix C gives a complementary full-support terminal reflection family in that rank, with maximum common descent set $\{1, 3, 5, 7, 9\}$ and the same non-cover vanishing conclusion (Proposition 14). Standard-parabolic embeddings transport the affine and rank-ten examples to larger E_n groups; these embedded families retain the support of their original parabolics.

6 Scope

Interval transfer. Assuming the combinatorial-invariance theorem of [17, Theorem 1.1], an interval isomorphic to a finite simply laced FC-bottom interval has $\mu \in \{0, 1\}$ by Corollary 2. The same observation transfers the uniform and fixed-rank bounds to intervals admitting those models, with $\mu = 0$ for an affine-family model. Indeed, the polynomial and relative rank are preserved. The target bottom need not be FC. This is a formal conditional consequence; no new family of cross-type interval isomorphisms is asserted.

Full polynomials. Vanishing of μ does not imply $P = 1$. In D_4 , let x be the product of the three leaves and let c be the center. The element $b = xcx$ has maximum common commuting descents and an even gap $\ell(b) - \ell(x) = 4$, but $P_{x,b} = 1 + 2q$ [13, §3, preprint Corollary 3]. Thus the distinction between the leading coefficient and the full polynomial is essential, even for the descent mechanism used here.

A Computational certificates

The terminal classification, the affine inversion count, and the affine FC-catalogue and cover checks are computational parts of the proofs. All arithmetic is exact. For E_6 and E_7 , one implementation stores simple-root images as root indices, and an independent implementation stores integer columns; neither imports the other. For E_8 , two different parabolic decompositions certify the same terminal list. The E_7 method uses 138,240 candidates at its final stage and 143,660 across the entire recursive chain. The independent D_7 method uses 280 right-terminals and 2,160 cosets, giving 604,800 candidates.

For the single inherited D_6 coefficient, the code also reproduces

$$P_{x_6, w_6}(q) = 1 + 6q + 11q^2 + 6q^3 + q^4 + q^5.$$

The verifier `computations/verify_certificate.py` checks all 24,245 dependency records in `results/d6-recurrence-certificate.json`, every correction term, all right-descent recurrences

at the target pair, and its R -polynomial reciprocity identity, without calling the KL evaluator. This confirms the cited value; the value itself is Gern’s result.

The supplementary FC catalogues use the recurrence

$$w \in W_{\text{FC}} \iff R(w) \text{ is commuting and } ws \in W_{\text{FC}} \text{ for every } s \in R(w). \quad (5)$$

Matsumoto’s theorem justifies the test: a noncommuting braid before the last letter survives in a shorter prefix, while a length-three braid ending at the last letter gives adjacent right descents. Closure under all ascents passing the test is exhaustive, since every FC element has FC weak predecessors. Bruhat comparison and both descent inclusions then check the eligible endpoint assertion independently of Corollary 8. These FC catalogues are supplementary for the finite theorem, but essential for excluding covers in the affine family of Theorem 13.

The D_7 cosets are obtained from the orbit of the integral fundamental weight

$$(4, 7, 10, 8, 6, 4, 2, 5).$$

Its simple-root pairings are 1 at node 0 and zero elsewhere. Distinct orbit images give distinct cosets; the orbit has the full index 2, 160, and the stored representatives have no right descent in D_7 . These checks establish completeness and minimality.

Check	E_6	E_7	E_8
Roots	72	126	240
FC elements	662	2,670	10,846
Noncommuting terminals	1	4	6
Commuting terminals	22	36	58
FC star moves checked	3,620	16,776	77,476

The full E_6 and E_7 searches contain 51,840 and 2,903,040 elements, with maximum lengths 36 and 63. Their length distributions agree with the Weyl-group Poincaré products. The FC counts agree with [16, §5.1]. These are consistency checks; exhaustiveness follows from the closure and pruning arguments in Section 3. An independent E_8 verifier checks the terminal words across both decompositions, lengths by all 120 positive-root inversions, both terminal tests, complete FC closure, and every eligible lower endpoint. It also checks the D_7 parabolic/coset Poincaré product against the E_8 exponents 1, 7, 11, 13, 17, 19, 23, 29.

The affine E_9 FC catalogue has 44,199 elements and maximum length 44. Its closure is certified by (5). Deleting one letter from the length-27 base word and retaining length-26 products gives exactly 21 distinct Bruhat covers, none fully commutative. Independent root matrices also verify all 240 affine inversion-string records used in Theorem 13.

The self-contained proof supplement is distributed as `exceptional-leading-proof.zip` in [GitHub release v0.1.0](#). It includes the manuscript, the proof sources and fixed inputs, an expected-output summary, and a SHA-256 manifest. The following paths are relative to its `payload/` directory; output certificates are generated in an isolated working copy by the runner.

Result	Programs and output certificates
E_6, E_7	research/verify_exceptional.py results/e6-independent-certificate.json results/e7-independent-certificate.json
E_8	research/en_e8/reproduce.py research/en_independent/verify_e8.py results/e8-independent-audit.json
Affine E_9	affine_proof.py research/en_affine_referee/verify_fc_catalogue.py research/en_affine_referee/fc-cover-certificate.json
Uniform family	research/en_uniform/referee_verify.py research/en_uniform/referee-certificate.json
E_{10}	research/en_affine_referee/verify_indefinite_e10.py research/en_affine_referee/e10-certificate.json

The uniform root proof is symbolic. Arithmetic checks in finitely many ranks are supplementary, and do not replace the all-rank argument. After extracting the archive, run `python3run_proofs.py` from its top-level directory, using Python 3.10 or later and an installed C++17 compiler. No external Python packages, network access, or package installation are required. The runner checks every input hash and then rebuilds the flat and recursive E_6, E_7, E_8 terminal classifications, the independent E_8/D_7 classification, and the needed FC catalogues. It verifies the uniform identities, affine family and cover exclusion, E_{10} family, and finite matching certificates. It also verifies the saved D_6 recurrence certificate; it does not regenerate that polynomial with a KL evaluator. The resulting mathematical summaries and terminal matrices must agree with the supplied expectations.

The default mode checks the older full-group E_6, E_7 snapshots against the regenerated parabolic classifications. The command `python3run_proofs.py--full` additionally regenerates those full-group root-index and integer-matrix searches. Assertions must remain enabled. Each run retains its working copy, commands, output certificates, and logs. The recursive engine shares root arithmetic with the flat engine; the D_7 search and integer-matrix verifiers are separate implementations. The proof-only entry point excludes random discovery searches and capped unresolved coefficient probes.

B Seed data

The matrix of the affine base word, with columns $b_0(\alpha_j)$, is

$$B_0 = \begin{pmatrix} 0 & 0 & 1 & -1 & 1 & 0 & 0 & 0 & -1 \\ 1 & 0 & 2 & -2 & 2 & -1 & 1 & -1 & -2 \\ 2 & -1 & 3 & -3 & 3 & -2 & 2 & -1 & -2 \\ 2 & -1 & 2 & -2 & 3 & -2 & 2 & -1 & -2 \\ 2 & -1 & 1 & -1 & 2 & -1 & 1 & -1 & -1 \\ 2 & -1 & 1 & -1 & 1 & 0 & 1 & -1 & -1 \\ 1 & -1 & 1 & 0 & 0 & 0 & 1 & -1 & -1 \\ 1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & -1 & 2 & -2 & 2 & -1 & 1 & -1 & -1 \end{pmatrix}.$$

For the E_{10} seed, successive simple reflections in the following order reduce β_0 to α_9 :

$$\begin{aligned} &1, 3, 5, 4, 7, 6, 5, 9, 2, 1, 0, 3, 2, 1, 4, 3, 2, \\ &5, 6, 7, 8, 9, 2, 1, 0, 3, 2, 1, 4, 3, 2, 5, 4, 3, 6, \\ &5, 4, 7, 6, 5, 9, 2, 1, 0, 3, 2, 1, 4, 3, 2. \end{aligned}$$

Each intermediate vector is nonnegative and its height decreases. For the uniform finite- E_8 root γ , the fixed sequence

$$2, 1, 0, 4, 3, 2, 1, 6, 5, 4, 3, 2$$

reduces it to the branch simple root. For $\gamma + \delta$, concatenate that sequence with

$$\begin{aligned} b, 2, 1, 0, 3, 2, 1, 4, 3, 2, 5, 4, 3, 6, 5, 4, 7, 6, 5, \\ b, 2, 1, 0, 3, 2, 1, 4, 3, 2, \end{aligned}$$

where b denotes the branch label ($4r$ or 9); the concatenated sequence again ends at the branch simple root. These fixed witnesses also certify the affine real-root input to Lemma 11.

C An indefinite E_{10} supplement

Embed the affine subdiagram on $\{0, \dots, 7, 9\}$ and take

$$\delta = (2, 4, 6, 5, 4, 3, 2, 1, 0, 3), \quad \gamma = (1, 2, 3, 2, 2, 1, 1, 0, 0, 1).$$

Here γ differs from the root used in the affine family of Section 5. Set

$$\beta_0 = (3, 7, 10, 9, 7, 6, 4, 3, 1, 6), \quad I = \{1, 3, 5, 7, 9\},$$

set $T = r_\gamma r_{\gamma+\delta}$ and, for integers $k \geq 0$, define

$$\beta_k = \beta_0 - k\gamma + (k^2 + 3k)\delta, \quad b_k = r_{\beta_k}. \quad (6)$$

Proposition 14. *For integers $k \geq 0$, the reflections in (6) are distinct full-support terminal elements of $W(E_{10})$, with $L(b_k) = R(b_k) = I$. For every fully commutative x , $\mu(x, b_k) \in \{0, 1\}$, and $\mu(x, b_k) = 0$ whenever $x < b_k$ and $\ell(b_k) - \ell(x) > 1$.*

Proof. The lowering word in Appendix B reduces β_0 to α_9 , proving real-root membership. The roots γ and $\gamma + \delta$ are real roots of the embedded affine parabolic, with $(\beta_0, \gamma) = -3$ and $(\beta_0, \delta) = -1$. Lemma 11 gives $\beta_k = T^k \beta_0$. The binomial increments $4\delta - \gamma$ and 2δ are nonnegative, so every coordinate of β_k is strictly positive. Its simple-root pairings are

$$\begin{aligned} ((\alpha_j, \beta_k))_{j=0}^9 = & (-1, 1, -2 - k, 1 + k, -1 - k, 1 + k, -1 - k, 1 + k, \\ & -1 - k^2 - 3k, 2 + k). \end{aligned}$$

The positive entries occur exactly on I , and their sums with adjacent entries are nonpositive. The reflection formula and (1) give both descent sets and terminality. The constant pairing $(\alpha_1, \beta_k) = 1$ makes column 1 change every coordinate row, proving full support. The coordinate $\beta_{k,0} = 2k^2 + 5k + 3$ proves distinctness. The set I is a maximum independent set of size 5. Reflection parity and Proposition 6 give the coefficient assertions. As in Theorem 3, the reflections are not FC because their (-1) -eigenspaces have dimension one, whereas that of $i(I)$ has dimension five. $\square$

The base reflection has length 101. No formula for the lengths of the indefinite families is needed. Standard-parabolic embeddings transport the E_9 and E_{10} families to larger generalized E_n groups; their supports remain those of the smaller parabolics.

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